Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 31)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
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7	
8	
9	
10	
11	
TOTAL	

Do not write outside the box

Answer all questions in the spaces provided.

0 1 . 1 Describe coal gasification and coal liquefaction and explain how each increases the usefulness of coal.

[4 marks]

0 1 . 2 Describe what methane hydrate is and where it is found.

[2 marks]

0 1 . 3 Describe **two** proposed methods of extracting methane from hydrate deposits and explain **two** environmental risks.

[4 marks]

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0 2 A student measured the mass of carbon stored per hectare in three land uses in the same region.

Table 1

Land use	Carbon in vegetation / t ha^{-1}	Carbon in soil dead organic matter / t ha^{-1}
Mature woodland	120	95
Permanent pasture	3	80
Arable (ploughed annually)	2	45

0 2 . 1 Calculate the total carbon lost per hectare when mature woodland is converted to arable farmland.

[1 mark]

Answer ☐ t/ha

0 2 . 2 Explain why the soil carbon in arable land is so much lower than in pasture, even though the vegetation carbon is similar.

[2 marks]

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0 2 . 3 Suggest **two** ways the arable farmer could increase soil carbon storage.

[2 marks]

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0 3 Figure 1 shows the daily deaths in London and the mean concentration of smoke in the air during the smog of December 1952.

Figure 1

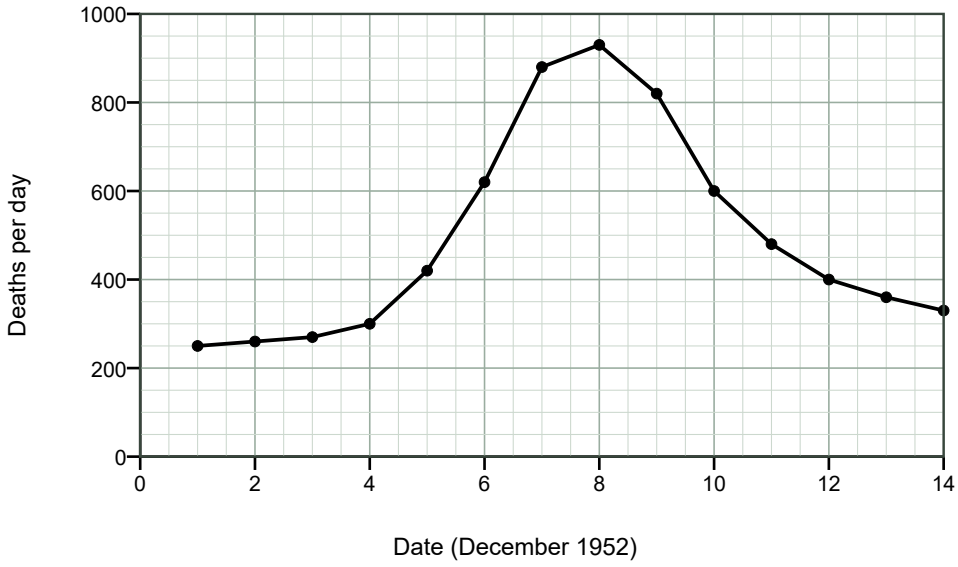


Table 1

Date	3 Dec	5 Dec	7 Dec	8 Dec	10 Dec	12 Dec
Smoke / mg m^{-3}	0.4	1.5	4.4	4.5	1.2	0.5

0 3 . 1 Describe the relationship between smoke concentration and daily deaths shown in Figure 1 and Table 1.

[2 marks]

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0 3 . 2 Calculate the percentage increase in daily deaths between 1 December and 8 December.

[2 marks]

Answer ..... %

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0 3 . 3 Explain how the weather conditions in December 1952 caused the smog to form.

[3 marks]

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0 3 . 4 Explain why most of the deaths were among the elderly, the very young and people with existing respiratory disease.

[2 marks]

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0 3 . 5 State the legislation introduced in the UK in response to this smog.

[1 mark]

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0 4 Table 1 shows measurements taken from a stream above and below the point where drainage water from an old metal-mine spoil heap enters it.

Table 1

Site	pH	Dissolved zinc / mg dm^{-3}	Dissolved oxygen / % saturation	Number of i nvertebrate taxa	Mayfly larvae present?
Upstream	7.1	0.02	96	18	Yes
50 m downstream	3.4	6.5	94	3	No
2 km downstream	5.9	1.1	95	9	No

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0 4 . 1 Explain how the acidic drainage water is produced in the spoil heap.

[3 marks]

0 4 . 2 Use Table 1 to explain why the number of invertebrate taxa falls so sharply 50 m downstream, even though dissolved oxygen is unchanged.

[3 marks]

0 4 . 3 Suggest why the pH and zinc concentration have partly recovered 2 km downstream.

[2 marks]

0 4 . 4 Describe how acid mine drainage can be controlled.

[2 marks]

Do not write outside the box

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0 5 Table 1 shows the bypass ratio and first year of use of jet engines used on four airliners.

Table 1

Engine	Aircraft	Bypass ratio	First use
RR Olympus	Concorde	0 : 1	Early 1950s
RR Conway	Boeing 707	0.25 : 1	Late 1950s
P&W JT9D	Boeing 747	5 : 1	1969
RR Trent 1000	Boeing 787	10 : 1	2011

0 5 . 1 Explain how a high bypass ratio reduces engine noise.

[3 marks]

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0 5 . 2 Describe **two** other features of aircraft design that reduce noise.

[2 marks]

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0 5 . 3 Explain how each of the following airport operating procedures reduces noise nuisance.

[4 marks]

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Constant descent angle .....

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Night flight restrictions .....

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Noise-based landing charges / quota count .....

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Multiple runways .....

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0 5 . 4 Explain why aircraft are noisier on take-off than on landing.

[1 mark]

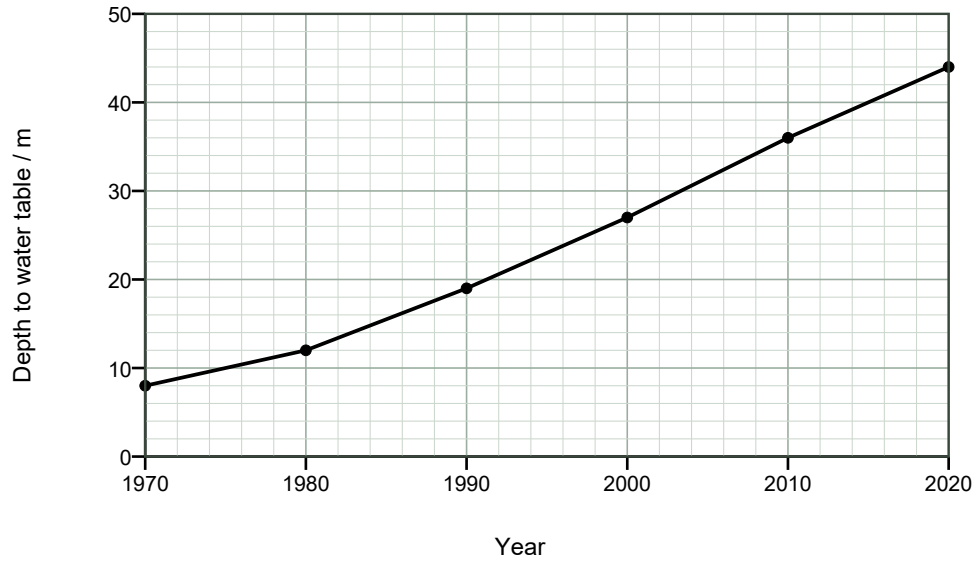
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0 6 Figure 1 shows the depth of the water table below the ground surface in an aquifer beneath a coastal farming region between 1970 and 2020.

Figure 1



0 6 . 1 Calculate the mean rate at which the water table fell between 1970 and 2020.

[1 mark]

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Answer _____ m/yr

0 6 . 2 Explain why the water table is falling.

[2 marks]

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0 6 . 3 Explain the features a rock must have to form an aquifer.

[3 marks]

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0 6 . 4 Describe **four** consequences of over-exploiting this coastal aquifer.

[4 marks]

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0 7 Table 1 shows the running mean of moth catches as a student increased the number of nights of light-trapping at one site.

Table 1

Number of nights	**1**	**2**	**4**	**6**	**8**	**10**	**12**	**14**
Running mean catch per night	31	18	24	21	22	22	21	22

0 7 . 1 Use Table 1 to suggest the minimum number of nights needed to obtain a reliable mean. Explain your answer.

[2 marks]

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0 7 . 2 Explain why moth surveys must be carried out at standardised times and weather conditions.

[3 marks]

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0 7 . 3 Distinguish between **accuracy** and **precision** of measurements.

[2 marks]

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0 7 . 4 Explain the importance of a null hypothesis and standardised techniques in a scientific investigation.

[3 marks]

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0 8 . 1 Explain why mineral extraction causes more land-use conflict than most other human activities.

[2 marks]

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0 8 . 2 Describe how each of the following environmental impacts of an operating mine can be reduced.

[5 marks]

Dust .....

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Noise .....

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Turbid drainage water .....

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Acidic leachate from spoil .....

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Loss of amenity .....

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0 8 . 3 Explain how spoil heaps can be made stable and describe what happened at Aberfan in 1966.

[3 marks]

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0 9 . 1 Explain how ionising radiation damages living cells.

[3 marks]

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0 9 . 2 Distinguish between acute and chronic effects of radiation exposure.

[2 marks]

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0 9 . 3 Distinguish between **exposure** and **contamination**.

[2 marks]

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0 9 . 4 A worker receives an absorbed dose of 0.002 Gy from alpha radiation.
Calculate the effective dose in sieverts.

[2 marks]

Answer _____ Sv

0 9 . 5 State what is measured by the becquerel.

[1 mark]

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1 0 . 1 Neonicotinoids are the most widely used group of insecticides. Explain how the properties of neonicotinoids affect their impact on bee populations and describe the methods used to reduce this impact.

[9 marks]

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Extra space

1 0 . 2 State **one** way in which pesticide use in agriculture may be reduced by using natural processes.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 To what extent can renewable energy resources replace fossil fuels in meeting the energy needs of a modern society?

[25 marks]

OR

1 1 . 2 Discuss the extent to which the study of ozone depletion illustrates how scientific investigation can lead to the successful solution of an environmental problem.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS